PRACTICAL 01 / BEGINNER LAB

Check a deployed Warp Route

Name: _____ Date: _____

Aim. Inspect two testnets and verify a working route without moving tokens.

Before you start. Use existing-route after completing Windows setup and .env configuration. This exercise is read-only.

Commands

```
npm.cmd run preflight
npm.cmd run warp:check
npm.cmd run verify
Get-Content .\deployments\verification.json
```

Expected result

Both chain IDs are correct, the CLI reports no violations, both remote routers are enrolled, and locked BSC collateral equals outstanding Sepolia synthetic supply when no message is in flight. No gas is spent by these checks.

Record your observations

Origin / destination: _____
Amount or setting: _____
Dispatch hash or verification time: _____
Message ID / result: _____

Quick questions

- Which chain ID is BSC testnet? Which one is Sepolia?
- Why does an RPC balance check not prove message delivery?

Use testnets only. Run from the selected kit project folder. Use npm.cmd. Amounts are in smallest units. Never record a private key or recovery phrase on this sheet.

PRACTICAL 02 / BEGINNER LAB

Bridge 1 GLD to Sepolia

Name: _____ Date: _____

Aim. Make a standard forward transfer.

Before you start. Use an already deployed route. The signer needs at least 1 GLD on BSC and native gas on both networks.

Commands

```
$A = @(
  '--origin','bsctestnet','--destination','sepolia',
  '--amount','1000000000000000000','--relay'
)
npm.cmd run warp:send -- @A
# Run the next command only after delivery.
npm.cmd run verify
```

Expected result

After delivery, BSC owner GLD decreases by 1, locked BSC GLD increases by 1, and the same owner receives 1 synthetic GLD on Sepolia. Record both message ID and transaction hashes.

Record your observations

Origin / destination: _____
Amount or setting: _____
Dispatch hash or verification time: _____
Message ID / result: _____

Quick questions

- Does the BSC ERC-20 total supply change?
- Is the collateral router address the token you import on BSC?

Use testnets only. Run from the selected kit project folder. Use npm.cmd. Amounts are in smallest units. Never record a private key or recovery phrase on this sheet.

PRACTICAL 03 / BEGINNER LAB

Reverse direction: Sepolia to BSC

Name: _____ Date: _____

Aim. Change the origin chain and return the token.

Before you start. Complete practical 02 first and confirm that 1 GLD arrived on Sepolia. Use the same funded signer.

Commands

```
$A = @(
  '--origin','sepolia','--destination','bsctestnet',
  '--amount','1000000000000000000','--relay'
)
npm.cmd run warp:send -- @A
# Wait for delivery, then verify.
npm.cmd run verify
```

Expected result

Sepolia burns 1 GLD and BSC releases 1 GLD. If you began with no other bridged tokens, both locked collateral and synthetic supply return to zero. If immediate self-relaying fails, use practical 08 on the existing dispatch.

Record your observations

Origin / destination: _____
Amount or setting: _____
Dispatch hash or verification time: _____
Message ID / result: _____

Quick questions

- What changed in the command compared with practical 02?
- Why must the Sepolia token balance exist before the return?

Use testnets only. Run from the selected kit project folder. Use npm.cmd. Amounts are in smallest units. Never record a private key or recovery phrase on this sheet.

PRACTICAL 04 / BEGINNER LAB

Bridge a fraction: 0.1 GLD

Name: _____ Date: _____

Aim. Practice 18-decimal amount conversion with a small transfer.

Before you start. The route is deployed and the signer holds BSC GLD. Compare with practical 02; only the amount changes.

Commands

```
$A = @(
  '--origin','bsctestnet','--destination','sepolia',
  '--amount','100000000000000000','--relay'
)
npm.cmd run warp:send -- @A
# After delivery:
npm.cmd run verify
```

Expected result

Exactly 0.1 GLD moves. If starting from zero bridged supply, BSC locked collateral and Sepolia total synthetic supply both become 0.1. Do not type --amount 0.1: the CLI expects an integer in smallest units.

Record your observations

Origin / destination: _____
Amount or setting: _____
Dispatch hash or verification time: _____
Message ID / result: _____

Quick questions

- Write 0.1 GLD multiplied by 10^18.
- Does sending one tenth of a token reduce gas by ten times?

Use testnets only. Run from the selected kit project folder. Use npm.cmd. Amounts are in smallest units. Never record a private key or recovery phrase on this sheet.

PRACTICAL 05 / BEGINNER LAB

Bridge 5 GLD

Name: _____ Date: _____

Aim. Change the transfer size and compare token amounts with gas usage.

Before you start. The signer needs at least 5 GLD on BSC plus native gas. Complete earlier pending transfers before starting.

Commands

```
$A = @(
  '--origin','bsctestnet','--destination','sepolia',
  '--amount','5000000000000000000','--relay'
)
npm.cmd run warp:send -- @A
# After delivery:
npm.cmd run verify
```

Expected result

BSC locked collateral and Sepolia synthetic supply increase by 5 GLD. Record balances before and after. Results are deltas, so add any already bridged amount from earlier practicals.

Record your observations

Origin / destination: _____
Amount or setting: _____
Dispatch hash or verification time: _____
Message ID / result: _____

Quick questions

- What integer represents 5 GLD?
- Which native asset pays the origin transaction gas?

Use testnets only. Run from the selected kit project folder. Use npm.cmd. Amounts are in smallest units. Never record a private key or recovery phrase on this sheet.

PRACTICAL 06 / BEGINNER LAB

Use a different recipient

Name: _____ Date: _____

Aim. Bridge 0.25 GLD to a second public address on Sepolia.

Before you start. Create or choose a second account you control. Copy only its public address. The original owner signs and pays gas; the recipient does not need to sign to receive.

Commands

```
$Recipient = 'REPLACE_WITH_SECOND_PUBLIC_ADDRESS'
$A = @(
  '--origin','bsctestnet','--destination','sepolia',
  '--amount','250000000000000000',
  '--recipient',$Recipient,'--relay'
)
npm.cmd run warp:send -- @A
# After delivery, check recipient on Sepolia explorer.
npm.cmd run verify
```

Expected result

The second address receives 0.25 GLD. The verification helper still reports the original owner balance, so check the recipient balance in its wallet or explorer. Total backing still matches. Returning those tokens later requires the recipient account and Sepolia gas; it is not part of this sheet.

Record your observations

Origin / destination: _____
Amount or setting: _____
Dispatch hash or verification time: _____
Message ID / result: _____

Quick questions

- Why does ownerGLD not increase on Sepolia here?
- Can you receive tokens without sharing your private key?

Use testnets only. Run from the selected kit project folder. Use npm.cmd. Amounts are in smallest units. Never record a private key or recovery phrase on this sheet.

PRACTICAL 07 / BEGINNER LAB

Two small transfers

Name: _____ Date: _____

Aim. Send 0.05 GLD twice and compare with one 0.1 transfer.

Before you start. The owner has BSC GLD. Wait for each transfer to deliver before submitting the next command.

Commands

```
$A = @(
  '--origin','bsctestnet','--destination','sepolia',
  '--amount','50000000000000000','--relay'
)
npm.cmd run warp:send -- @A
# Confirm delivery before the second send.
npm.cmd run warp:send -- @A
# Confirm delivery again, then inspect totals.
npm.cmd run verify
```

Expected result

The combined amount is 0.1 GLD but there are two distinct dispatches and message IDs. Expect more transaction overhead than a single transfer of the same total. If either command fails after dispatch, check that message before continuing.

Record your observations

Origin / destination: _____
Amount or setting: _____
Dispatch hash or verification time: _____
Message ID / result: _____

Quick questions

- How many unique message IDs should you record?
- Is a second send a valid way to retry delivery of the first?

Use testnets only. Run from the selected kit project folder. Use npm.cmd. Amounts are in smallest units. Never record a private key or recovery phrase on this sheet.

PRACTICAL 08 / BEGINNER LAB

Inspect and retry delivery

Name: _____ Date: _____

Aim. Track an existing message without sending more tokens.

Before you start. Use a dispatch transaction hash from an earlier practical. The example is Sepolia-origin; change it to bsctestnet for a BSC-origin dispatch.

Commands

```
$Tx = 'REPLACE_WITH_EXISTING_DISPATCH_TX_HASH'
$A = @('--origin','sepolia','--dispatchTx',$Tx)
npm.cmd run status -- @A
# ONLY if not delivered, after checkpoint availability:
npm.cmd run status -- @A --relay
# After confirmed delivery:
npm.cmd run verify
```

Expected result

An already delivered message reports its delivery transaction, and you stop. A pending message can be retried when metadata is ready. You are relaying the existing dispatch, not issuing another token transfer. Do not deliberately break security settings to simulate an outage.

Record your observations

Origin / destination: _____
Amount or setting: _____
Dispatch hash or verification time: _____
Message ID / result: _____

Quick questions

- Which hash is needed: approval, dispatch, or delivery?
- Why can validator metadata take time to become available?

Use testnets only. Run from the selected kit project folder. Use npm.cmd. Amounts are in smallest units. Never record a private key or recovery phrase on this sheet.

PRACTICAL 09 / BEGINNER LAB

Switch the BSC RPC endpoint

Name: _____ Date: _____

Aim. Use an alternate RPC while keeping the same contracts and chains.

Before you start. No message should be in flight. This is a read-only practical: do not deploy or send tokens.

Commands

```
notepad.exe .env
# Change this ONE setting in Notepad and save:
# BSC_TESTNET_RPC=https://bsc-testnet.drpc.org
npm.cmd run preflight
npm.cmd run warp:check
npm.cmd run verify
# Optional: restore the original URL and recheck.
```

Expected result

The alternate RPC must report chain 97. Token/router addresses and balances should agree with the original endpoint, allowing for block timing. If the endpoint is unavailable, record that result and restore the known working URL; do not switch to mainnet.

Record your observations

Origin / destination: _____
Amount or setting: _____
Dispatch hash or verification time: _____
Message ID / result: _____

Quick questions

- Does changing RPC deploy a new token?
- Why is the chain ID check necessary before using a new URL?

Use testnets only. Run from the selected kit project folder. Use npm.cmd. Amounts are in smallest units. Never record a private key or recovery phrase on this sheet.